


# DATABASE MANAGEMENT SYSTEMS

## *GROUP ASSIGNMENT*

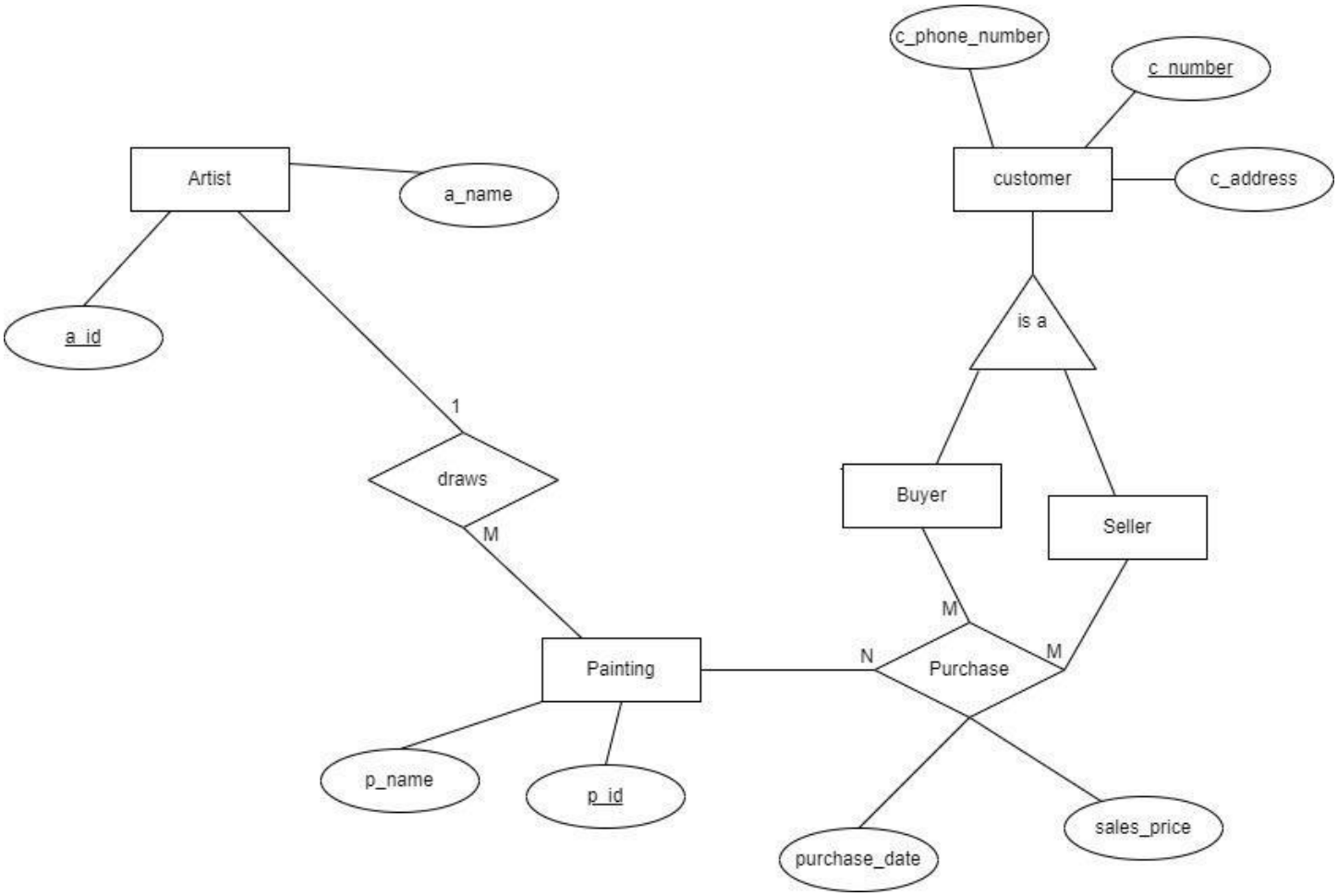
Group No.: Group 18
Batch No.: 23.1

## QUESTION 01

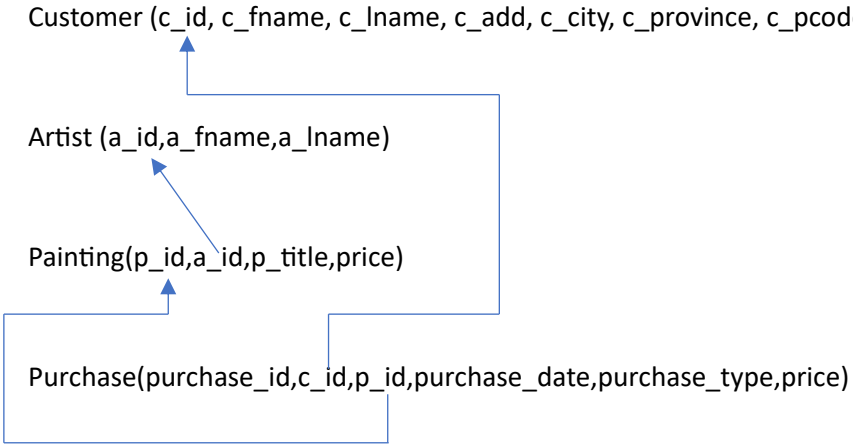
Assumptions: Single painting can have many buyers and many sellers

A single buyer can purchase many paintings and a seller can sell many paintings

A single artist can draw many paintings a single painting can only have one artist



## Question 02


Customer (c_id, c_fname, c_lname, c_add, c_city, c_province, c_pcode, c_phone)
Artist (a_id,a_fname,a_lname)
Painting(p_id,a_id,p_title,price)
Purchase(purchase_id,c_id,p_id,purchase_date,purchase_type,price)

## QUESTION 03

```
MariaDB [(none)]> create database fineartdb;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> use fineartdb;
Database changed
MariaDB [fineartdb]>
```

```
MariaDB [fineartdb]> CREATE TABLE Customer(
    -> CusID int NOT NULL PRIMARY KEY,
    -> firstName varchar(255),
    -> lastName varchar(255),
    -> CusAdd varchar(255),
    -> CusCity varchar(255),
    -> CusProvince varchar(255),
    -> CusPostalCode varchar(255),
    -> CusPhone varchar(255));
Query OK, 0 rows affected (0.019 sec)
```

```
MariaDB [fineartdb]> CREATE TABLE Artist (
    ->     ArtistID INT PRIMARY KEY,
    ->     ArtistName VARCHAR(100));
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [fineartdb]> CREATE TABLE Painting (
    -> PaintingID int NOT NULL PRIMARY KEY,
    -> ArtistID INT,
    -> PaintingTitle varchar(255),
    -> Price REAL,
    -> FOREIGN KEY (ArtistID) REFERENCES Artist(ArtistID));
Query OK, 0 rows affected (0.043 sec)
```

```
MariaDB [fineartdb]> CREATE TABLE Purchase(
    ->  CusID int NOT NULL,
    ->  PaintingID int NOT NULL,
    -> PurchaseDate DATE,
    -> PurchaseType varchar(10),
    -> Price Real);
Query OK, 0 rows affected (0.022 sec)
```

```
MariaDB [fineartdb]> ALTER TABLE purchase
    -> ADD PRIMARY KEY(CusID,PaintingID);
Query OK, 0 rows affected (0.129 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [fineartdb]> ALTER TABLE Purchase
    -> ADD FOREIGN KEY (CusID) REFERENCES Customer(CusID);
Query OK, 0 rows affected (0.069 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [fineartdb]> ALTER TABLE Purchase
    -> ADD FOREIGN KEY (PaintingID) REFERENCES Painting(PaintingID);
Query OK, 0 rows affected (0.061 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

**QUESTION 04 (a)**

```
MariaDB [fineartdb]> INSERT INTO Customer values
    -> (1, 'Meth', 'Wijesundara', 'Address 1', 'Kadawatha', 'WP', '12345', '123-456-7890'),
    ->     (2, 'Ashini', 'Reshana', 'Address 2', 'Matara', 'SP', '56789', '555-123-4567'),
    ->     (3, 'Nadeel', 'Ifthika', 'Address 3', 'Kelaniya', 'WP', '98765', '888-888-8888'),
    ->     (4, 'Damsith', 'Dissanayake', 'Address 4', 'Nugegoda', 'WP', '54321', '999-999-9999'),
    ->     (5, 'Ashen', 'Suri', 'Address 5', 'Kandy', 'UP', '23456', '111-222-3333');
Query OK, 5 rows affected (0.013 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

**(b)**

```
MariaDB [fineartdb]> select * from customer where cusID<5;
+-------+-----------+-------------+-----------+-----------+-------------+---------------+--------------+
| CusID | firstName | lastName    | CusAdd    | CusCity   | CusProvince | CusPostalCode | CusPhone     |
+-------+-----------+-------------+-----------+-----------+-------------+---------------+--------------+
|     1 | Meth      | Wijesundara | Address 1 | Kadawatha | WP          | 12345         | 123-456-7890 |
|     2 | Ashini    | Reshana     | Address 2 | Matara    | SP          | 56789         | 555-123-4567 |
|     3 | Nadeel    | Ifthika     | Address 3 | Kelaniya  | WP          | 98765         | 888-888-8888 |
|     4 | Damsith   | Dissanayake | Address 4 | Nugegoda  | WP          | 54321         | 999-999-9999 |
+-------+-----------+-------------+-----------+-----------+-------------+---------------+--------------+
4 rows in set (0.001 sec)
```

**(c)**

```
MariaDB [fineartdb]> INSERT INTO Artist
    -> values
    -> (3, 'Caroline Channing'),
    -> (15, 'Ben Frings');
Query OK, 2 rows affected (0.013 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

**(d)**

```
MariaDB [fineartdb]> delete from artist
    -> where ArtistName =  "Dennis Fringes";
Query OK, 0 rows affected (0.001 sec)
```

**(e)**